Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

AI Smart Doorbell

Eoin Fitzpatrick | B.Eng. (Hons) Software & Electronic Engineering | Final Year Project

React + Vite

FastAPI + WebSocket

SQLite + ML Pipeline

Project Overview

AI Smart Doorbell is an assistive system designed to help older people particularly those living with Alzheimer’s or memory loss recognise who is at their door. A live webcam feed is processed in real time to identify known visitors such as family members or carers, displaying their name clearly on screen. Users can also respond using simple hand gestures to simulate opening or locking the door, with all data stored privately on a local device.

Key Deliverables

Real-time face recognition displayed in the browser

Simple gesture control — thumbs up to open, palm to lock

Local enrolment and storage, no cloud or internet required

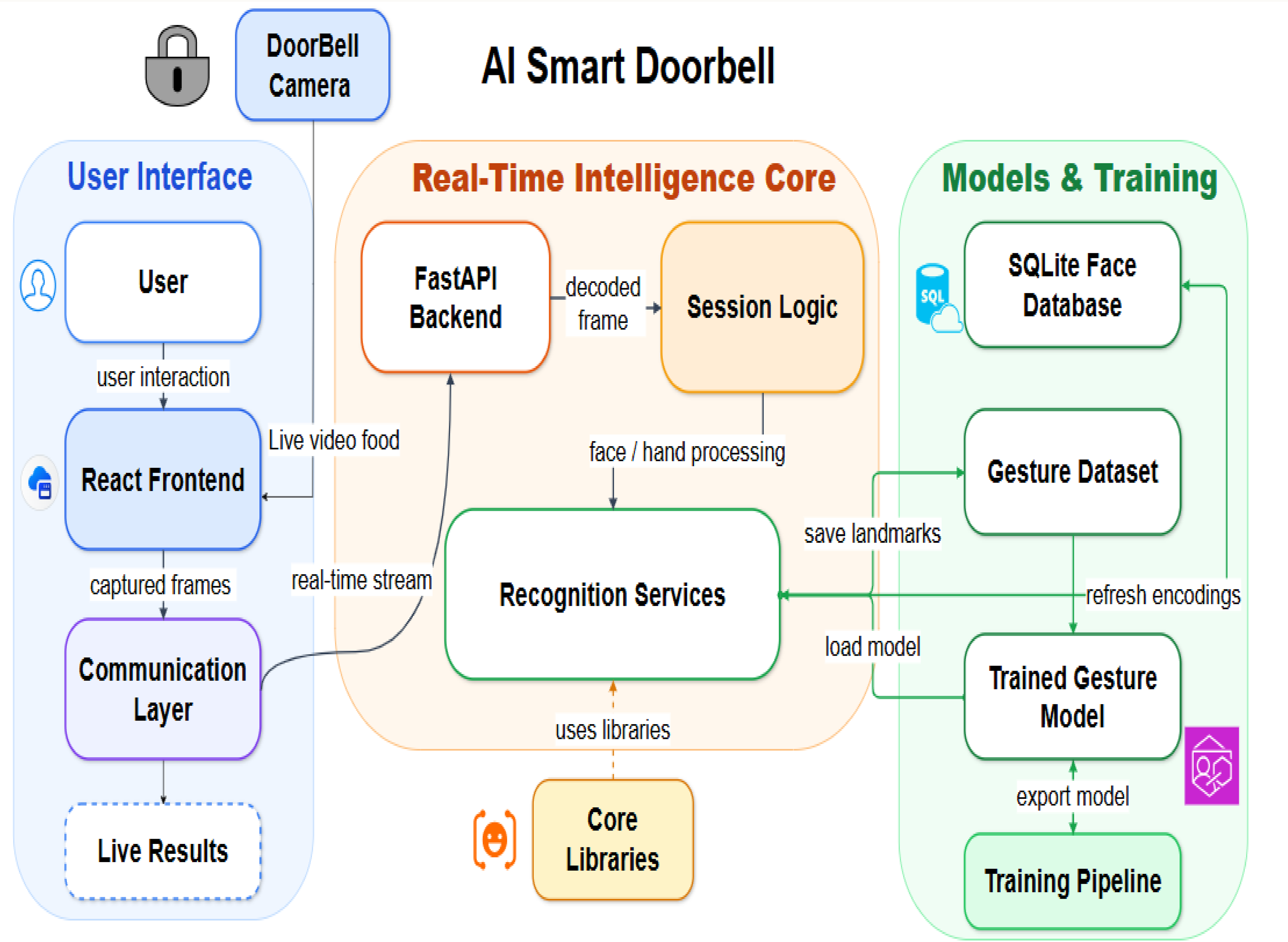
Technical Highlights

All face and visitor data stored locally no cloud or internet required.

Each visitor session is handled independently for reliable, accurate results.

Gesture model trained and validated on 513 samples with 100% test accuracy.

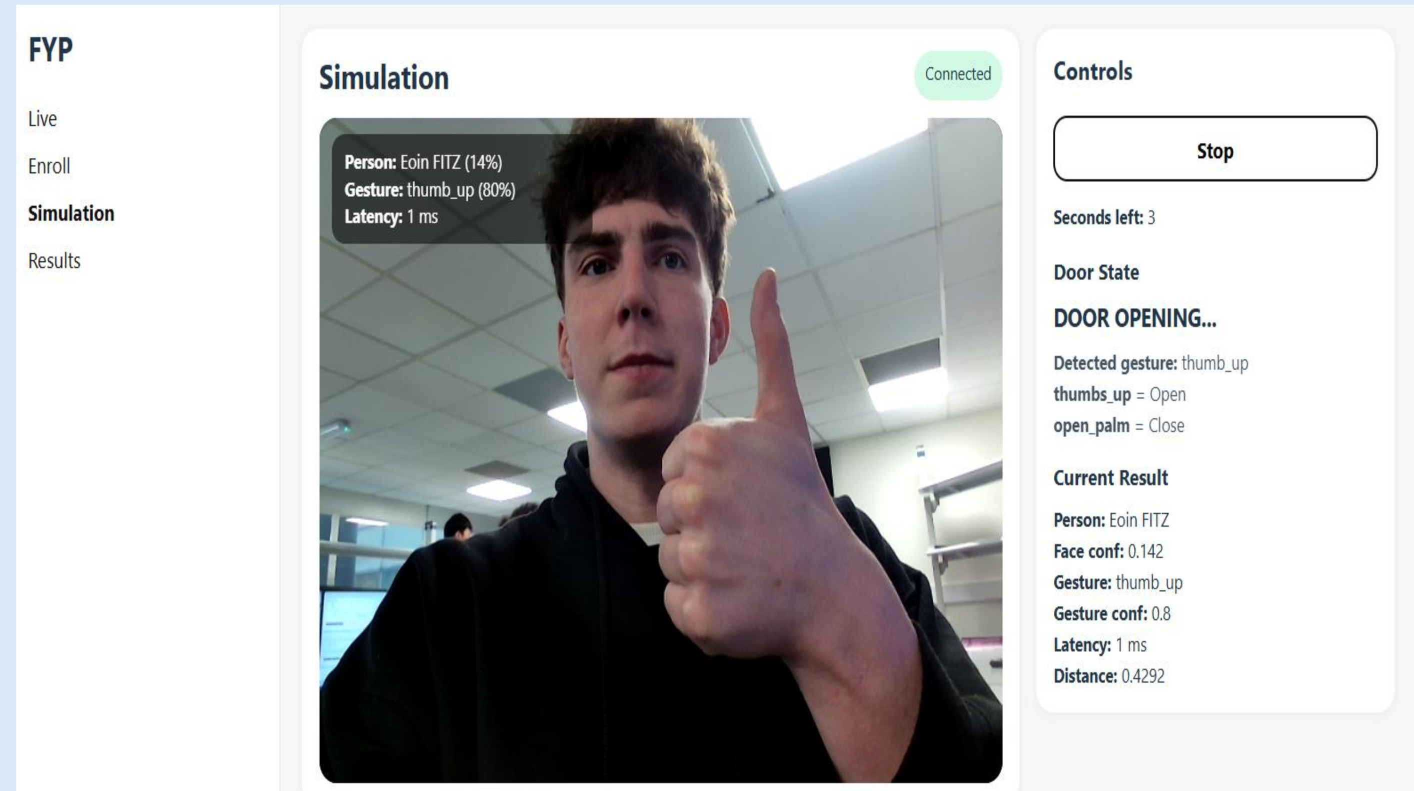
Architecture Diagram



Measured Results from the Repository

100% Gesture accuracy (2 classes)	**200** Trees in gesture classifier	**64** Hand landmark features	**103** Test samples	**80ms** Local WebSocket latency

User Experience Modes



Enroll

Add known visitors like family or carers stored privately on device

Name: ________

Samples: 10

Simulation

Use a hand gesture to open or lock the door for a recognised visitor

DOOR OPENING

Conclusion and Future work

All project objectives were successfully met, delivering a real-time assistive doorbell system proven to support older people with memory difficulties.

Future Work: mobile app with visitor notifications and integration with a real smart lock.